

José Giral Barajas

PHD STUDENT · MATHEMATICS RESEARCH

Imperial College London, South Kensington, London, SW7 2AZ, UK

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Education

Imperial College London

PH.D. MATHEMATICS

- Advisor: Professor Paul C Bressloff

London, UK

2024 - present

Universidad Nacional Autónoma de México (UNAM)

M.Sc. MATHEMATICS

- Advisor: Dr. Sergio I. López Ortega
- Thesis: Probabilistic epidemiological model for heterogeneous populations
- Qualifying exams in partial differential equations, probability and stochastic processes, and graph theory.

Mexico City, Mexico

2022 - 2024

Universidad Nacional Autónoma de México (UNAM)

B.Sc. APPLIED MATHEMATICS

- Advisor: Dr. Sergio I. López Ortega
- Thesis: Probabilistic epidemiological model with microdynamics and reverse time (Spanish)
- First-class honours, ranked top of class, and awarded the Gabino Barreda medal

Mexico City, Mexico

2017 - 2021

Professional Experience

2021-2022 **Business Intelligence Specialist**, Totalplay Telecomunicaciones, S.A. de C.V., Mexico City, Mexico

Publications

PREPRINTS

1. **J. Giral-Barajas**, S. Linn and P. C. Bressloff. Universality in the target arrival statistics of non-conservative search processes arXiv preprint: 2606.16025 (2026).

JOURNAL PAPERS

2. **J. Giral-Barajas** and P. C. Bressloff. Resetting-induced instability in queues fed by a search process in an interval *Phys. Rev. E* **113** 064101(2026).
3. **J. Giral-Barajas** and P. C. Bressloff. Stochastic 1D search-and-capture as a G/M/c queueing model. *J. Phys. A: Math. Theor* **58** 355001 (2025).
4. **J. Giral-Barajas**, C. I. Herrera-Nolasco, M. A. Herrera-Valdez and S. I. López. A probabilistic approach for the study of epidemiological dynamics of infectious diseases: Basic model and properties. *J. Theor. Biol.* **572** 111576 (2023).

Scholarships and Awards

SCHOLARSHIPS

2024 **Roth Scholarship for Ph.D. Degree**, Department of Mathematics, Imperial College London, UK

2022 **Graduate Scholarship for M.Sc. Degree**, National Council of Humanities, Sciences and Technologies (CONAHCYT), Mexico

UNIVERSITY AWARDS

2025 **PhD Symposium Award**, Imperial College London, UK

2022 **Gabino Barreda medal**, Universidad Nacional Autónoma de México (UNAM), Mexico

TRAVEL AWARDS

2026 **SIAM Student Travel Award**, To attend the SIAM Conference of the Life Sciences 2026

2026 **Doris Chen Mobility Award**, Imperial College London, UK

Teaching Experience

Spring 2026 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics

Fall 2025 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics

Spring 2025 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics

Spring 2024 **UNAM**, Graduate Teaching Assistant, Stochastic processes

Fall 2023 **UNAM**, Teaching Assistant, Probability I

Spring 2022 **UNAM**, Teaching Assistant, Mathematical Modelling Workshop

Posters

Jun 2025 Department of Mathematics PhD Symposium Poster Competition (1st prize), *Imperial College London, London, UK*

Nov 2023 Evento Inaugural de la Red Mexicana de Biología & Matemáticas, *UNAM, Cuernavaca, Mexico*

Talks and Workshops

CONTRIBUTED AND INVITED TALKS

Aug 2026 Advances in Asymptotic Analysis and Numerical Methods for First Passage Time Problems in Biology, *University of Notre Dame, Kylemore Abbey, Ireland*

Jul 2026 14th European Conference on Mathematical & Theoretical Biology (contributed), *University of Graz, Graz, Austria*

Jul 2026 SIAM Conference on the Life Sciences 2026 (invited), *SIAM, Cleveland, U.S.*

Jun 2026 Probability and Stochastic Processes seminar (invited), *Universidad Nacional Autónoma de México, Mexico City, Mexico*

Jun 2026 Mathematical Biology seminar (invited), *Universidad Nacional Autónoma de México, Mexico City, Mexico*

Jun 2026 Workshop on Extreme Events in Biological Function (Lightning talk), *National Institute for Theory and Mathematics in Biology, Chicago, USA*

Feb 2026 Online Conference on Population Dynamics, Ecology and Evolution (contributed), *Society for Mathematical Biology, Online*

Nov 2025 SIAM Postgraduate Conference 2025 (contributed), *University of Bristol, Bristol, UK*

Jun 2025 End-of-year Biomath Symposium (contributed), *Imperial College London, London, UK*

Dec 2024 Non-equilibrium Systems Group seminar (contributed), *Imperial College London, London, UK*

WORKSHOPS ATTENDED

Aug 2026 Advances in Asymptotic Analysis and Numerical Methods for First Passage Time Problems in Biology, *University of Notre Dame, Kylemore Abbey, Ireland*

Jun 2026 NITMB Workshop on Extreme Events in Biological Function, *National Institute for Theory and Mathematics in Biology, Chicago, USA*

Jan 2026 4th London Workshop on Active Matter, *Imperial College London, London, UK*

Outreach

Jul 2026	Co-organizer of SIAM Conference on the Life Sciences 2026 Minisymposium, Minisymposium on Stochastic Transport and Resource Accumulation in Biology, <i>Co-organized with Dr. Samantha Linn</i>
Oct 2025 - present	Member of the SIAM Student chapter committee, Student chapter at Imperial College London, <i>Imperial College London, London, UK</i>
Oct 2025	Imperial Natural Sciences blog post, Wandering searchers and supermarket queues: mathematically exploring how cells move resources, <i>Imperial College London, London, UK</i>
Aug 2025	Research Fest, Modelling Accumulation of Resources Queueing Theory and First-Passage Processes, <i>Imperial College London, London, UK</i>
Mar 2019	Science Fair Workshop, Cómo calcular Pi con balines (General public), <i>UNAM, Mexico City,</i> <i>Mexico</i>